

Will Donovan

E-mail: donovan@mail.tsinghua.edu.cn
Teams: dr.will.donovan
Homepage: w-donovan.github.io



Academic positions

2021– Yau MSC, Tsinghua University: Associate Professor
2018–21 Yau MSC, Tsinghua University: Assistant Professor
2014–18 Kavli IPMU, University of Tokyo: Postdoctoral Fellow
2011–14 University of Edinburgh: Research Assistant

Further affiliation

2021– Yanqi Lake BIMSA: Adjunct Associate Professor
2020– Kavli IPMU, University of Tokyo: Visiting Scientist

Education

2007–11 PhD Mathematics, Imperial College London (with R.P. Thomas, E. Segal)
2003–4 MMath Mathematics, University of Cambridge (Part III)
2000–3 BA Mathematics, University of Cambridge

Publications

Papers

1. Grassmannian twists on the derived category via spherical functors
Proc. London Math. Soc. (2013) 107 (5): 1053–1090.
2. Window shifts, flop equivalences and Grassmannian twists (with E. Segal)
Compositio Math. (2014) 150 (6): 942–978.
3. Mixed braid group actions from deformations of surface singularities (with E. Segal)
Comm. Math. Phys. (2015) 335 (1): 497–543.
4. The Pfaffian-Grassmannian equivalence revisited (with N. Addington and E. Segal)
Alg. Geom., Foundation Compositio Math. (2015) 2 (3): 332–364.
5. Noncommutative deformations and flops (with M. Wemyss)
Duke Math. Jour. (2016) 165 (8): 1397–1474.
6. Moduli spaces of torsion sheaves on K3 surfaces (with N. Addington and C. Meachan)
J. London Math. Soc. (2016) 93 (3): 846–865.
7. Mukai flops and $\mathbb{P}$ -twists (with N. Addington and C. Meachan)
J. Reine Angew. Math. (Crelle) (2019) 748: 227–240.

8. Contractions and deformations (with M. Wemyss)
Amer. J. Math. (2019) 141 (3): 563–592.
9. Twists and braids for general 3-fold flops (with M. Wemyss)
J. Eur. Math. Soc. (2019) 21 (6): 1641–1701.
10. Perverse schobers and wall crossing
Int. Math. Res. Notices (2019) 18: 5777–5810.
11. Noncommutative enhancements of contractions (with M. Wemyss)
Adv. Math. (2019) 344: 99–136.
12. Perverse schobers on Riemann surfaces: constructions and examples
Eur. J. Math. (2019) 5: 771–797.
13. Mirror symmetry for perverse schobers from birational geometry (with T. Kuwagaki 桑垣樹)
Comm. Math. Phys. (2021) 381: 453–490.
14. Stringy Kähler moduli for the Pfaffian-Grassmannian correspondence
SIGMA (2021) 17, 028.
15. Relating derived equivalences for simplices of higher-dimensional flops
Adv. Stud. Pure Math. (2023) 88: 305–330.
16. Root stacks and periodic decompositions (with A. Bodzenta)
Manuscripta Math. (2024) 175: 53–73.
17. Stringy Kähler moduli, mutation and monodromy (with M. Wemyss)
J. Diff. Geom. (2025) 129 (1): 115–164.
18. Derived symmetries for crepant contractions to hypersurfaces
Preprint, arXiv:2409.19555.
19. Window categories for a simple 9-fold flop of Grassmannian type
(with W. Hara, M. Kapustka, M. Rampazzo)
Preprint, arXiv:2510.06184.
20. Quiver braid group action for a 3-fold crepant resolution (with Luyu Zheng 郑璐予)
Preprint, arXiv:2512.19140.
21. Higher-dimensional conifolds and hypergeometric systems (with Weilin Su 苏蔚琳)
In preparation.
22. Categorical crepant contractions and spherical functors
In preparation.

Proceedings

23. Grassmannian twists, derived equivalences and brane transport
Proc. Symp. Pure Math. (2015), Proc. String-Math 2012.
24. Contractions of 3-folds: deformations and invariants
Internat. J. Math. (2016) 27 (7), special issue VBAC 2014.
25. Applications of noncommutative deformations
Proc. Kinoshita Alg. Geom. Symp. (2016), available online.

Grants

2019– China central government **Thousand Talents** Plan
Value: 2.5 million yuan (~\$350k)

2016–18 Japan Society for Promotion of Science KAKENHI **Young Scientist B**
Value: 2.9 million yen (~\$25k)

Awards

2020 Yau MSC Ruolin 若琳 **Research Excellence** Award

2019 Yau MSC **Outstanding Paper** Award ‘Twists and braids for general 3-fold flops’

Postgraduate students

Doctoral

2021–26 **Nantao Zhang** 张南涛 birational and enumerative geometry, Bridgeland stability
‘Stability conditions on local Calabi–Yau fourfolds and on blowups’
Postdoctoral researcher, Sun Yat-Sen University (2026–)

2020–25 **Luyu Zheng** 郑璐予 toric geometry, braid groups, perverse sheaves
‘Braid groups associated with exceptional surfaces in 3-folds’
Postdoctoral researcher, SCMS, Fudan University (2026–)

2019–23 **Xun Lin** 林汛 derived categories, birational geometry, Hodge theory
‘Non-commutative aspects of Hodge conjecture, minimality, and Torelli problems’
Assistant professor, CUHK Shenzhen (2026–)
Postdoctoral fellow, HKUST & MPIM Bonn (2023–26)

Masters

2019–25 **Weilin Su** 苏蔚琳 categorification, hypergeometric systems, mirror symmetry
‘Derived monodromy for the Appell F_4 hypergeometric system’

Examining

2024 **Caroline Namanya** doctoral external examiner
‘Pure braids and group actions from 3-fold flops’
Makerere University, Uganda / Glasgow University, UK

Mentoring

Junior faculty

2023–25 **Beihui Yuan** 袁北慧
Assistant professor, BIMSA, Beijing (2023–)

Postdocs

2026– **Marco Rampazzo**

2024–26 **Alicia Lamarche**
Assistant professor, BIMSA, Beijing (2026–)

2024–26 **Aimeric Malter**
Assistant professor, HIMIS, Shenzhen (2026–)

Teaching

Yau MSC, Tsinghua University

2026	Topics in derived categories
2026	Theoretical mechanics
2025	Topics in complex algebraic geometry
2025	Topics in algebraic geometry (homological mirror symmetry)
2024	Theoretical mechanics
2023	Topics in algebraic geometry (derived categories and toric geometry)
2023	Calculus 2
2022	Theoretical mechanics
2022	Lie groups and Lie algebras
2021	Geometric representation theory 2 (with Penghui Li 李鹏辉)
2021	Derived functors in algebraic and birational geometry
2020	Calculus 1
2020	Mirror symmetry: categories and constructions
2019	Geometric representation theory 2 (with Peng Shan 单凡)
2019	Calculus 2
2018	Derived functors in algebraic and birational geometry

Yanqi Lake BIMS

2023	Introduction to topos quantum theory
2022	Quiver approaches to machine learning
2022	Categorical and geometric aspects of quantum mechanics

Service

Organization

- Summer school ‘Perverse sheaves and their categorification in geometry and representation theory’ at Venice International University, organizer (2026)
- Workshop ‘Homological algebra of the infrared’ at TSIMF, Sanya, organizer (2023)
~20 speakers, including A. Bondal, M. Kapranov, Fields medallist M. Kontsevich
- Workshop ‘Derived categories’ at Hausdorff Institute, Bonn, organizer (2014)

Yau MSC, Tsinghua

- Institution Representative, China-Africa Mathematics Center (2026–)
- Foreign affairs committee (2023–)
- Tutor, Qiuzhen college (2023–)
- Events Hub organizer (2021–)
- Geometric representation theory seminar organizer (2021–)
- Colloquium committee (2019–21)

Mini Courses

2027	Makerere U, Uganda	‘Crepant contractions and spherical functors’
2025	Yonsei U, Seoul	‘Derived symmetries, crepant contractions and noncommutative deformations’
2020	KIAS, Seoul	‘Perverse schobers and mirror symmetry’
2012	Steklov Inst, Moscow	‘Symmetry and the derived category’

Selected talks

2026	2019 (continued)	2015
ICBS, Beijing	Australian National U	Nagoya U
IMS, CUHK, Hong Kong	BIRS Oaxaca, Mexico	Fields Institute, Toronto
2025	CTU, Prague	Max Planck Inst, Bonn
Cetraro, Italy	Fudan U, Shanghai	Oberwolfach
Wuhan U, China	Kyoto U	Warwick U
2024	Liverpool U	2014
Fujiyoshida, Japan	Osaka U	Freie Universität, Berlin
Makerere U, Uganda	Sydney U	BICMR, Beijing
UBC, Vancouver	SYSU, Guangzhou	Hamburg U
TSIMF Sanya, Hainan	2018	KIAS, Seoul
2023	Hokkaido U	Sheffield U
IMPAN, Warsaw	Nagoya U	2013
Kavli IPMU, Tokyo	TSIMF Sanya, Hainan	Cardiff U
Krakow, Poland	2017	Columbia U
Max Planck Inst, Bonn	Australian National U	Duke U
2022	Bath U	Edinburgh U
Cambridge U	ICMS, Edinburgh U	Imperial College London
Oberwolfach	Nagoya U	Leeds U
UCL, London	Osaka U	Oxford U
2021	SISSA, Trieste	Schrödinger Inst, Vienna
BICMR, Peking U	2016	2012
Cardiff U	BIRS Banff	ETH Zürich
CMSA, Harvard U	ECM, Berlin	Glasgow U
2020	Freie Universität, Berlin	Hausdorff Institute, Bonn
IASM, Hangzhou	Hanga Roa, Chile	Newcastle U
KAIST, Korea	Hong Kong U	Queen Mary, U London
Steklov Inst, Moscow	Kinosaki Symposium	Warwick U
2019	Kyoto U	
Auckland U, New Zealand	METU, Ankara	

Extended visits

Apr–May 2024 Sydney Mathematical Research Institute
Mar 2024 IMS, Chinese University of Hong Kong
Jun–Jul 2020 Korean Institute of Advanced Study, Seoul
Feb–Jun 2020 Kavli IPMU, University of Tokyo
Jan–Apr 2014 Hausdorff Institute, Bonn
Apr–Jul 2013 Schrödinger Institute, Vienna
May 2012 Steklov Institute, Moscow

Outreach

- Funckup night ‘Always be calculating’, Beijing (2026)
- Workshop series, for University of Tokyo ‘Soap bubbles and spacetime’ (2017–)
At 6 locations, including Tamarokuto Science Centre and Kavli IPMU
- Interviewed, for University of Tokyo ‘Discover Our People’ (2016)
- Talk: ‘The frontiers of symmetry: hidden geometry and tangled superstrings’
King’s College London Math School (2015)
Summerhall, Edinburgh, for International Year of Crystallography (2014)

Previous employment

2006–7 Lead programmer, Bloomberg LP: financial research and development
2005 TEFL teacher, Kyoto, Japan

Languages

- Japanese (JLPT N2), Mandarin (advanced)
- English (native)